A Creative Weak Supervised Semantic Segmentation for Remote Sensing Images

Zhibao Wang, *Member, IEEE*, Huan Chang, Lu Bai, *Member, IEEE*, Liangfu Chen, and Xiuli Bi

***Abstract*— In weakly supervised semantic segmentation (WSSS) tasks on remote sensing images, it is a common practice to train a classification network from scratch using a large batch of images with a limited number of classes. Subsequently, class activation maps are extracted from the model based on predefined class indices, and these maps are then optimized to obtain pseudolabels. To make this strategy effective when introducing a new class, a substantial amount of data needs to be provided to the model. In this article, we present an innovative framework, RS-TextWS-Seg, designed to efficiently generate high-quality segmentation results for a wide range of remote sensing objects using concise descriptions. Our proposed framework comprises three sequential stages: initially, we undertake parameter fine-tuning of the contrastive language-image pretraining (CLIP) model to swiftly strengthen its capacity for zero-shot detection of a limited number of remote sensing features. Subsequently, we introduce a text-driven background suppression mechanism aimed at deriving class activation maps from the refined CLIP model based on textual cues, while concurrently mitigating background noises. Finally, we use the segment anything model (SAM) to refine the edges of the extracted class activation map. We widely researched the leading-edge methodologies in WSSS and conducted a range of comparative experiments and ablation studies to prove the efficacy of our proposed framework. The research findings underscore that RS-TextWS-Seg outperforms other state-of-the-art methods on renowned datasets such as DLRSD and Potsdam, as well as on bespoke datasets specifically curated for overground petroleum pipelines and oil well fields.**

***Index Terms*— Fine-tuning, remote sensing image, text prompts, weakly supervised semantic segmentation (WSSS).**

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Zhibao Wang and Huan Chang are with the Department of Bohai Rim Energy Research Institute, Northeast Petroleum University, Qinhuangdao 066004, China, and also with the School of Computer and Information Technology, Northeast Petroleum University, Daqing 163318, China (e-mail: wangzhibao@nepu.edu.cn; 228003071186@stu.nepu.edu.cn).

Lu Bai is with the School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast, BT9 5BN Belfast, U.K. (e-mail: l.bai@qub.ac.uk).

Liangfu Chen is with the Department of State Key Laboratory of Remote Sensing Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: chenlf@radi.ac.cn).

Xiuli Bi is with the Department of Computer Science and Technology, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: bixl@cqupt.edu.cn).

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I. Introduction

HARNESSING semantic segmentation to derive information such as area, length, and quantity of various remote sensing objects from remote sensing images is a task of paramount importance. It propels advancements in sectors such as urban planning, agricultural monitoring, and resource management. Natural image datasets such as PASCAL VOC [1] and COCO [2] have rich volume and category diversity, meeting the requirements of countless pixel level supervised semantic segmentation experiments. Conversely, in the field of remote sensing, the effectiveness of pixel-level supervised semantic segmentation methods is limited by the small number of categories and quantities in datasets annotated by remote sensing experts. Moreover, the annotation of remote sensing semantic segmentation datasets not only demands costly human labor but also extensive periods, making it increasingly challenging to swiftly fulfill the diverse application needs and urban decision-making processes. Given the accessibility of image-level labels, weakly supervised semantic segmentation (WSSS) leveraging remote sensing imagery has gained considerable interest and acclaim.

The baseline for image-level WSSS of remote sensing images [3], [4], [5], [6], [7], [8], [9] involves training an image classification network, extracting class activation maps for specified categories from the model [10], and subsequently optimizing these maps to obtain pseudolabels. These methodologies have effectively enhanced the accuracy of target localization, and to a certain extent, have facilitated class activation mappings (CAMs) in achieving more comprehensive coverage of the intended targets. However, models developed from scratch through these approaches tend to demonstrate better performance only on a limited range of remote sensing objects.

A viable solution to this challenge involves using the existing knowledge within models to rapidly generate pseudolabels. To harness textual information for pseudolabel acquisition, we can integrate the pretrained contrastive language-image pretraining (CLIP) model as a classification network. To adapt to multiple categories, several approaches [11], [12], [13] leverage CLIP's zero-shot capability to generate pseudolabels. However, they share a common challenge: CLIP's zero-shot performance on remote sensing images is not optimal [14]. To address this, Liu et al. [15] proposed RemoteCLIP, which organizes existing remote sensing datasets through data augmentation and continues training on the CLIP model. While

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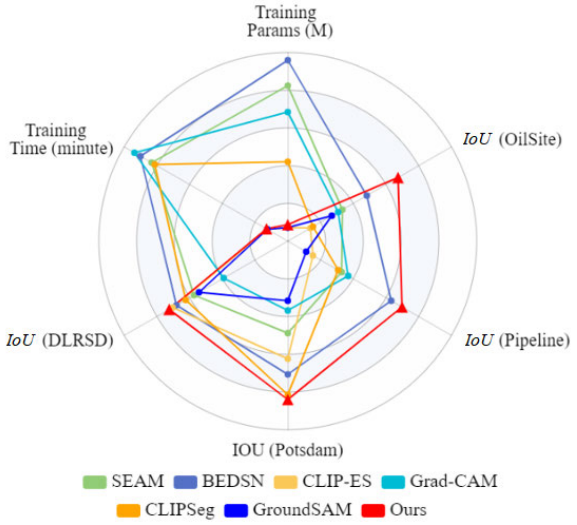


Fig. 1. Performance of each method and average IoU on each dataset.

this approach is effective, it heavily relies on large existing remote sensing datasets, making it ineffective when encountering small amounts of unknown land cover categories in remote sensing images. Therefore, efficient fine-tuning of CLIP to adapt to remote sensing images is urgently needed.

The introduction of the RS-TextWS-Seg framework marks a significant step forward in addressing these challenges. In the initial phase, fine-tuning of the weight parameters of the CLIP model is conducted. Subsequently, in the following stage, the text-based background suppression mechanism (TBBS Mechanism) is introduced in order to facilitate the extraction of a basic binary image from the refined CLIP architecture. In the third phase, the model link module (ML Module) is introduced to seamlessly integrate CLIP with segment anything model (SAM). Illustrated in Fig. 1, RS-TextWS-Seg showcases remarkable efficacy across a range of four different datasets.

Our contributions are summarized as follows.

1) We have introduced a streamlined and rapid WSSS framework built upon fine-tuning the CLIP model. Our research has demonstrated that fine-tuning CLIP with a small dataset of image-text pairs can effectively enhance its zero-shot performance on remote sensing imagery.
2) Our innovation includes the TBBS Mechanism, which not only extracts foreground targets based on brief textual descriptions but also reduces potential background noise. By incorporating this mechanism with the fine-tuned CLIP model, we are able to swiftly and accurately derive a basic CAM, establishing a meaningful linkage between human visual perception of objects and their semantically corresponding textual representations.
3) Additionally, we propose the integration of the ML Module to intelligently incorporating CLIP with SAM. This fusion enables the prior knowledge from CLIP to be transmitted to SAM in the form of points or bounding boxes, facilitating SAM to generate finely edged masks with precision.

The remainder of this study is organized as follows: Section II reviews related work to this research. Section III provides a detailed description of the proposed RS-TextWS-Seg framework. In Section IV, the experimental results and analysis are presented. Finally, conclusions are drawn in Section V.

II. Related Works

Within this section, our focus will initially include a retrospective analysis of the most recent image-level WSSS techniques applied to both natural and remote sensing imagery. Furthermore, we will briefly discuss the crucial support provided by the CLIP model and SAM model.

A. WSSS Based on Image-Level Labels

To obtain precise pseudolabels from complex remote sensing images, scholars have adeptly created modules to refine CAMs, yielding more comprehensive pseudolabels. Alternatively, they have integrated multilevel features for pinpointing targets with greater accuracy or have innovatively proposed background suppression strategies to reduce noise in pseudolabels. Recognizing that CAMs derived from the initial classifier often prove incomplete, Fang et al. [4] introduced an adversarial climbing (AC) strategy. This iterative approach perturbs pixel gradients in high-resolution remote sensing images, thereby elevating classification scores for constructing detailed class activation maps, specifically for the building class. Yan et al. [5] employed a combination of pixel-level affinity augmentation (PAA) and consistency regularization, ensuring the generation of consistent and precise CAMs. In a distinctive approach, Yan et al. [7] harnessed the MSG module to integrate global semantic information, guiding multilevel feature learning across different hierarchies. Subsequently, they employed multiple layers of features to generate multiscale CAMs, amplifying the extraction of valuable information from diverse hierarchical levels. Yan et al. [9] introduced an innovative the semantic perception-based multilevel progressive aggregation (SMPA) module, enhancing the integration of multilevel features. Additionally, Zhang et al. [6] executed subtraction operations on fused foreground and background saliency maps, effectively reducing background noise within pseudolabels. Qiao et al. [16] integrated the spatial correlation refinement (SCR) module into the encoder–decoder network, adeptly suppressing noise. The unified shortcomings of the aforementioned solutions include two main aspects. First, weakly supervised networks trained from scratch require prolonged training time and a substantial volume of remote sensing datasets. Second, the limitation arises from the occurrence of multiple classes of targets mixed within the class activation maps. Utilizing existing knowledge from pretrained models can effectively alleviate the first issue. Meanwhile, combining image-level labels with text and employing text for precise localization of remote sensing objects within images can to some extent address the second problem.

Zhou et al. [10] have introduced the language-guided segmentation network (WLSegNet), a novel language-guided segmentation approach. This method achieves open-vocabulary

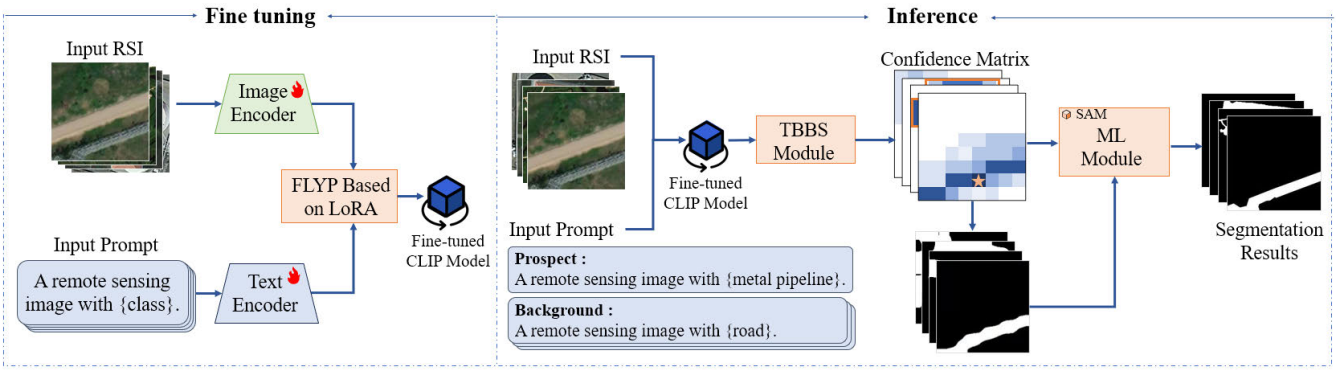


Fig. 2. Flowchart of RS-TextWS-Seg generating segmentation results. During the fine-tuning phase, the weights of the CLIP model are adjusted using paired images and text descriptions to obtain a fine-tuned model. In the inference stage, both images and their corresponding foreground and background text descriptions serve as inputs. These inputs are processed through the TBBS module and ML module, resulting in a segmented image.

semantic segmentation by learning generalized context vectors with batch aggregation (mean) and mapping class prompts to image features through a frozen CLIP model [17]. Lin et al. [12] used a CLIP model, CLIP, for category localization without further training, relying solely on image-level labels. Lüddecke and Ecker [13], utilizing CLIP as the backbone network, trained an extended version of the PhraseCut dataset. Employing a transformer-based decoder, they obtained binary segmentation maps for additional image generation based on free-form textual prompts or expressive queries. More recently, Zou et al. [18], building upon SAM, proposed a prompt and interactive model called segment everything everywhere all at once (SEEM). This model excels in simultaneously segmenting everything in an image under textual guidance. ZegFormer [19] and SimSeg [20] pioneer a two-step process, initially employing MaskFormer [21] to learn the generation of mask proposals for classes with uncertainties. Subsequently, these proposals undergo classification using the knowledge derived from a pretrained visual-linguistic model such as CLIP [14]. Presenting an innovative advancement, ZegCLIP [22] introduces a single-stage methodology, seamlessly extending CLIP's zero-shot prediction capability directly from the image level to the pixel level. Nevertheless, despite their commendable performance across diverse domains, these methodologies exhibit suboptimal outcomes when applied to remote sensing imagery, primarily due to the scarcity of remote sensing images within the pretraining datasets.

B. CLIP

CLIP [14] has exhibited significant success and potential in various visual tasks within a zero-shot environment. It comprises an image encoder and a text encoder. In a batch of image-text pairs, the CLIP model learns embeddings to gauge the similarity between images and text. Trained on an extensive dataset consisting of 400 million image-text pairs, CLIP surpasses fixed sets in smaller datasets such as PASCAL VOC2012 [1]. It demonstrates the capability to recognize a broader and more diverse range of object categories. It is noteworthy that CLIP exhibits relative weakness in some specialized, complex, or abstract tasks, such as satellite image classification (EuroSAT and RESISC45), lymph node tumor detection (PatchCamelyon), counting objects in computer-generated scenes (CLEVRCounts), and tasks related to autonomous driving [14]. CLIP is employed in various downstream tasks such as semantic segmentation [12], [13], video understanding [23], linking 3-D point clouds with language [24], video question answering [25], and so forth. Depending on the specific scenarios encountered in each task, CLIP is often fine-tuned or further trained to achieve better performance.

C. Segment Anything (SAM)

SAM [26] exhibits remarkable proficiency in generating high-fidelity pseudolabels. Comprising three integral components—an image encoder, a prompt encoder, and a mask decoder. The image encoder computes embeddings for images, while the prompt encoder embeds various prompts such as points, boxes, scribblings, and masks. Subsequently, these rich sources of information are seamlessly fused within a mask decoder, which proficiently predicts segmentation masks. SAM's efficacy is assessed on the SA-1B dataset, boasting an extensive collection of over 1 billion masks corresponding to more than 11 million images [26]. The evaluation highlights SAM's exceptional performance, particularly in the challenging zero-shot setting. Recently, there have been many research works utilizing SAM. For example, Grounding-DINO [27] utilizes SAM to achieve replacement and deletion of target regions after segmentation. SAM-CLIP [28] inherits both SAM and CLIP models, and integrates multitask learning, continual learning techniques, and teacher–student knowledge distillation.

III. Methods

In this section, we will present the innovative RS-TextWS-Seg framework that we have developed. Initially, we create a selection of remote sensing dataset pairs in an image-text format to fine-tune the parameters of the CLIP model. Subsequently, we extract pseudolabels from the fine-tuned CLIP model, as demonstrated in Fig. 2. By inputting remote sensing images along with a set of relevant foreground and background texts, we leverage the gradient-weighted class activation map (Grad-CAM) [29] method to yield a confidence matrix. This matrix is then refined through the TBBS module to generate a preliminary pseudolabel. To enhance this pseudolabel further,

we utilize the coordinates of points or bounding boxes in conjunction with the remote sensing images as inputs to the SAM module, based on the number of connected regions within the confidence matrix. Finally, employing the ML module, we derive the refined pseudolabel. Detailed insights into each stage will be provided in Sections III-A–III-C.

A. Fine-Tuning Method Like You Pretrain (FLYP) Based on Low-Rank Adaptation (LoRA)

The FLYP fine-tuning method involves the conversion of downstream class labels into textual prompts, followed by the ongoing optimization of the contrastive loss between image embeddings and prompt embeddings describing the classes (referred to as contrastive fine-tuning). Previous methods of fine-tuning [30], [31] have been subtle modifications to the standard supervised training paradigm, aimed at enhancing classification performance by minimizing cross-entropy loss on the image classifier. In this study [32], conducted on CLIP ViT-B16, averaging across seven out-of-distribution (OOD) datasets (comprising two from WILDS [33] and five related to ImageNet classes), FLYP demonstrates a 4.2% increase in OOD performance over fully fine-tuned models. Moreover, it outperforms the current state-of-the-art linear probing then full fine-tuning (LP-FT) [30] approach by more than 1% in both in-distribution (ID) and OOD scenarios. These results highlight the applicability of this advantage in few-shot fine-tuning scenarios where each class has very few instances, proving the superiority of the FLYP fine-tuning method over previous approaches. When fine-tuning CLIP with FLYP, we employ class labels to construct prompts, enabling the direct minimization of the contrastive loss between these prompts and the image embeddings derived from the fine-tuning dataset in use.

Consider an image classification setting where the goal is to map an image $I \in \mathcal{I}$ to a label $y \in \mathcal{Y}$. We use image-text pretrained models like CLIP that learn joint embeddings of image and text. Let $f : \mathcal{T} \mapsto \mathbb{R}^d$ denote the image encoder that maps an image to a d-dimensional image-text embedding space. f is parameterized by the parameters θ_{img}. Let T be the space for text descriptions of images. Analogously, $g : \mathcal{T} \mapsto \mathbb{R}^d$ is the language encoder with model parameters θ_{text}. The backbone of the pretraining objective is contrastive learning, where the goal is to align the embedding $f(I_i)$ of an image close to the embedding $g(T_i)$ of its corresponding text description, and away from other text embeddings $g(T_i)$ in the batch. Given a batch with N images with their corresponding text descriptions $D = \{(I_i, T_i), \ldots, (I_N, T_N)\}$, pretraining objective is as follows:

$$\mathcal{L}_{\text{pre}}(D, \theta) := \sum_{i=1}^{N} -\log \frac{\exp\left(\overline{f}(I_i) \cdot \overline{g}(T_i)\right)}{\sum_{j=1}^{N} \exp\left(\overline{f}(I_i) \cdot \overline{g}(T_j)\right)} + \sum_{i=1}^{N} -\log \frac{\exp\left(\overline{f}(I_i) \cdot \overline{g}(T_i)\right)}{\sum_{j=1}^{N} \exp\left(\overline{f}(I_j) \cdot \overline{g}(T_i)\right)} \quad (1)$$

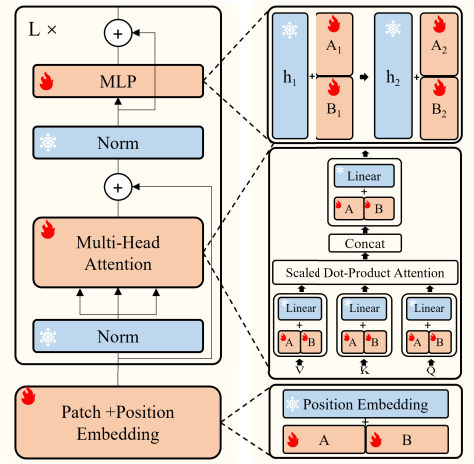


Fig. 3. Architecture of the proposed fine-tuned CLIP. It uses LoRA to impose constraints on the parameters of the CLIP model during the fine-tuning process and add parallel LoRA A and LoRA B layers in the connection layer of MLP, k, q, v in MHSA, and some layers in the patch embedding layer, respectively.

where $\theta = [\theta_{\text{img}}, \theta_{\text{text}}]$ are image and text encoder parameters, and $\overline{f}$ and $\overline{g}$ are the ℓ_2 normalized f and g.

Given downstream training samples $\{(x_1, y_2), \ldots, (x_n, y_n)\} \backsim P_{id}$. Since the pretrained image embeddings are trained to be aligned with the text embeddings, we can perform zero-shot classification without updating any weights. Given k classes (names) $\{c_1, c_2, \ldots, c_k\}$, we construct corresponding text descriptions $\{T_1, T_2, \ldots, T_k\}$ using templates (e.g., "A remote sensing image with c_1"). The zero-shot prediction corresponding to image I is $\arg\max_i \overline{g}(T_i)^\top \overline{f}(I)$, where $\overline{g}$ and $\overline{f}$ are the normalized text and image embeddings. This can be written as $\arg\max_i (h_{zs}^\top \overline{f}(I))_i$ where $h_{zs} \in \mathbb{R}^{\text{dk}}$ is the zero-shot linear head with columns corresponding to text descriptions of the classes T_k.

Expanding on the framework established by FLYP, we implemented subtle adjustments to the structure of the CLIP model, as illustrated in Fig. 3. This involved the incorporation of supplementary parallel layers within key units like multilayer perceptron (MLP) and multihead self-attention (MHSA), while leveraging the LoRA method to govern the interaction between these randomly initialized layers and the layers retaining the preexisting trained weights. During deployment, the CLIP model, fine-tuned with the LoRA method, aligns trainable matrices alongside frozen weight values in parallel. Considering a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, we prescribe its update by expressing it through a low-rank decomposition $W_0 + \Delta W f(x) = W_0 f(x) + \text{BA}$, where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and the rank $r \ll \min(d, k)$. During training, W_0 remains frozen and impervious to gradient updates, while A and B house trainable parameters. It is noteworthy that both W_0 and $\Delta W = \text{BA}$ are multiplied by the same input, and their respective output vectors are sum-reckoned coordinate-wise. For $h = W_0 f(x)$, our modified forward pass unfolds as follows:

$$h = W_0 f(x) + \Delta W_f(x) = W_0 f(x) + BAf(x). \quad (2)$$

We initiate the training process with a random Gaussian initialization for A and zero initialization for B, ensuring that

$\Delta W = BA$ is zero at the outset of training. Subsequently, we scale $\Delta W_f(x)$ by (α/r), where α is a constant related to r. When optimizing with Adam, fine-tuning α aligns roughly with tuning the learning rate, provided we appropriately scale the initialization. The encoder of CLIP [14] demonstrates versatility by accommodating Vision Transformers and ResNets models, while LoRA [34] is compatible with both Transformer and CNN architectures. Based on these insights, we have the flexibility to fine-tune various components of the CLIP model, including attention weights, MLP layers, LayerNorm layers, or other positional encoding layers, using LoRA. Illustrated in Fig. 3, we uphold the integrity of the original pretrained weight matrices in CLIP, making adjustments solely to the values of matrices A and B during the fine-tuning process. Given a label y, let $\mathcal{T}_y$ denote a set of possible text descriptions of the class. For example, these descriptions could include different contexts such as "A remote sensing image with {class}." Let $P_{\text{text}}(\cdot \mid y)$ denote the uniform distribution over text descriptions. Given a batch of labeled samples $D = \{(x_1, y_1), \ldots, (x_B, x_B)\}$, we construct a corresponding batch D' of image-text pairs and update the model parameters via stochastic gradient descent on the same pretraining objective (1). We summarize this in Algorithm 1. Inference using the finetuned encoders f' and g' is performed in the same way as zero-shot prediction, except using the finetuned encoders f' and g'. Precisely, the prediction for an image I is again given by $\arg\max_i \overline{g'}(T_i)^\top \overline{f'}(I)$.

Algorithm 1 Contrastive Finetuning Based on LoRA (One Batch)

Input:

Pretrained parameters W_0 of the target modules,
Labeled batch $D = \{(x_1, y_1), \ldots (x_N, x_N)\}$,
Distribution over text descriptions $P_{text}(\cdot \mid y)$, $y \in \mathcal{Y}$,
Scaling factor α and rank r in LoRA constraints,

Training Step:

1. Adds trainable pairs of rank decomposition matrices A and B in parallel to existing weight matrices W_0,
2. Create text/image paired data via labels $D = \{(I_1, T_1), \ldots (I_N, T_N)\}$, where $T_i \backsim P_{text}(y_i)$,
3. Update parameters via contrastive loss $W := W - \mathcal{L}_{pre}(D', W_0 + BA)$

Fine-tuning offers the advantages of shortened training durations and minimized image consumption. The reduction in training time is attributed to the constrained number of trainable parameters. In our experimental setup, following the introduction of parallel layers, we computed the quantity of trainable parameters using the following equation, where the trainable parameters are contingent upon the rank (r) and the inherent shape of the original weights

$$|\Theta| = 2 \times d_{\text{model}} \times r \times \hat{L}_{\text{LoRA}} \tag{3}$$

where $\hat{L}_{\text{LoRA}}$ is the number of weight matrices we apply LoRA to and d_{model} is the dimension of self-attention modules W_q, W_v, W_k, or W_o.

B. Extracting Segmentation Results Through TBBS

Grad-CAM [29] utilizes gradients flowing through the network to generate a coarse localization map for both CNNs and Transformer networks with multimodal inputs, highlighting crucial regions in the image used for predicting concepts. This can be achieved without the need to alter the architecture [29]. The weights assigned to each feature map for a specific class can be computed using the following equation:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial Y^c}{\partial A_{ij}^k}. \tag{4}$$

Then, the CAM map of class c at spatial location (i, j) can be obtained by (4). ReLU is adopted to ignore features that negatively influence the target class

$$\text{CAM}_{ij}^c = \text{ReLU}\left(\sum_k \alpha_k^c A_{ij}^k\right). \tag{5}$$

This article introduces a softmax function into Grad-CAM, enforcing mutual exclusivity among different categories. This transformation converts the original scores (logits) of Grad-CAM into a probability distribution. In a more specific way, the ultimate scores are derived by employing the softmax function, and the process is outlined as follows:

$$s^c = \frac{\exp(Y^c)}{\sum_{c'=1}^{C} \exp\left(Y^{c'}\right)}. \tag{6}$$

s^c represents the score assigned to the cth class following the application of softmax. Subsequently, these processed scores are utilized for gradient computation, enabling the calculation of class-specific feature weights as follows:

$$\begin{aligned} \alpha_k^c &= \frac{1}{Z} \sum_i \sum_j \sum_{c'} \frac{\partial Y^{c'}}{\partial A_{ij}^k} * \frac{\partial s^c}{\partial Y^{c'}} \\ &= \frac{1}{Z} \sum_i \sum_j \frac{\partial Y^c}{\partial A_{ij}^k} * s^c\left(1 - s^c\right) \\ &\quad + \frac{1}{Z} \sum_i \sum_j \sum_{c' \neq c} \frac{\partial Y^{c'}}{\partial A_{ij}^k} * s^c\left(-s^{c'}\right). \end{aligned} \tag{7}$$

The aforementioned (7) illustrates the attenuation of target feature map weights by nontarget classes, facilitating the fine-tuning of the target class's CAM via alternative classes. In order to segregate the pixels of the target class from those of the background class, we introduce a TBBS Mechanism. The entire process of this mechanism is depicted as shown in Fig. 4(a). This mechanism operates on the premise of delineating M categories based on the dataset, with other categories serving as the contextual backdrop when identifying the target category. Each potential category present in remote sensing imagery is briefly described in a prescribed format: "A remote sensing image with {class}." The model computes the relevance of M descriptions to an image and employs gradient backpropagation on the descriptor of the first category. This process yields a binary confidence matrix for the first category, representing the target class. Throughout this procedure, the fine-tuned model exhibits varying sensitivity to distinct features for each category. By simply adjusting the target

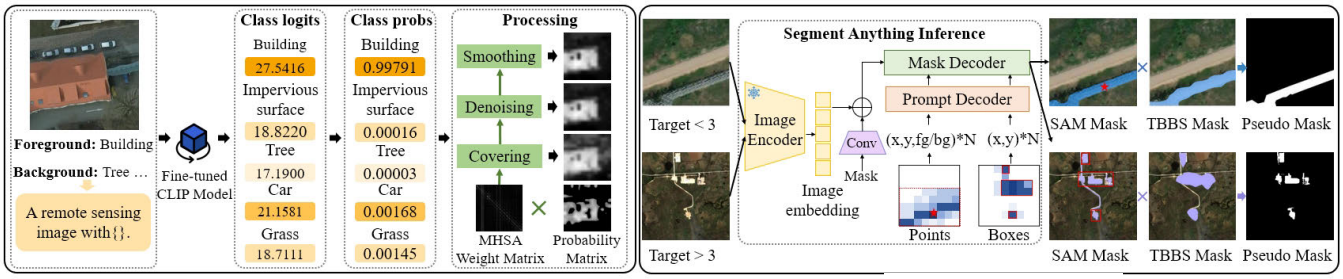


Fig. 4. TBBS module and ML module of RS-TextWS-Seg. (a) TBBS module: extract the foreground category activation map with the most prominent features through text, while suppressing noise from other background categories. Then, the MHSA weight matrices are used to increase the correct coverage area of the class activation graph, and finally, segmentation results are optimized through denoising and smoothing. (b) ML module: reasonably utilize SAM. When the connected area is not less than 3, use box to obtain segmentation results; otherwise, use point to obtain segmentation results.

and background categories, we effectively suppress the occurrence of specified categories within the background descriptor. This methodology diverges from previous approaches that necessitate retraining the classification network. Using CLIP's zero-shot capability and Grad-CAM's gradient backpropagation, this technique enables versatile application across multiple classes without incurring additional time investments

$$A = \frac{D + D^T}{2}, \text{where } D = \text{Sinkhorn}\left(W^{\text{attn}}\right). \tag{8}$$

The binary matrix of confidence $M^c \in \mathbb{R}^{h \times w}$ obtained from prior operations contains the most acute local features of the target. Leveraging the multihead attention mechanism enables the model to collectively focus on diverse positional information across distinct representational subspaces [35]. Within the refinement process, the MHSA extends pixel coverage toward semantically akin regions, thereby amplifying the matrix's scope over target land features. We compute the mean of eight sets of MHSA weight matrices $W^{\text{attn}} \in \mathbb{R}^{\text{hw} \times \text{hw}}$ to derive a novel attention weight matrix. Filtering this matrix using a predefined threshold λ facilitates denoising, yielding a binary representation. To obtain seamlessly contiguous pseudolabels, we employ Sinkhorn [36] normalization, rendering both rows and columns of the matrix into a transformed state, known as matrix D. Consequently, this process yields a new attention weight matrix, denoted as matrix A. By engaging in a more sophisticated level of probabilistic smoothing and denoising operations, refined pseudolabels are obtained. This process is as proven by (8).

C. Optimization Module for Pseudolabeling Based on SAM

SAM's architecture comprises a sophisticated ensemble—a large-scale image encoder, a prompt encoder, and a lightweight mask decoder. A comprehensive understanding of SAM's intricate model specifications can be found in the referenced literature on SAM [26]. To derive segmentation outcomes for specified objects, users must input points or bounding boxes as prompts into SAM's prompt encoder [37]. Notably, SAM currently lacks support for textual inputs as prompts. While SAM demonstrates a remarkable ability to produce sharply defined mask boundaries and maintain high segmentation efficiency, users are urged to supply SAM with accurate prior knowledge to attain precise segmentation outcomes. Recent reports have firmly established SAM as an exemplary tool for pseudolabel generation [38], showcasing a notable rise in mean intersection over union (MIoU) when effectively amalgamated with various weakly supervised methods [39]. The process of SAM inference is described in Algorithm 2.

Algorithm 2 Segment Anything Inference

Input:

Image $I \in \mathbb{R}^{3 \times H \times W}$,

$P_{point} \in \mathbb{R}^{N \times 2}$, P_{point} indicates that the input prompt are points, where N is the number of points and 2 represents the coordinates,

$B_{box} \in \mathbb{R}^{N*2}$, B_{box} represents that the input prompt are boxes, where N is the number of boxes and 2 represents the coordinates of the boxes.

Inference Step:

1. Input I into the image encoder to extract features and obtain image embeddings: $f^I = ViT(I)$, $f \in \mathbb{R}^{c \times h \times w}$,
2. Obtain dense encoding: $f^D \in \mathbb{R}^{c \times h \times w}$.
3. Obtain sparse encoding: $f^S \in \mathbb{R}^{K \times c}$,

For P_{point}, perform position encoding to obtain $f^{P_{point}} \in \mathbb{R}^{N \times c}$, where each point is mapped to a c-dimensional vector,

For B_{box}, first reshape it into two points, then encode the points in the same way as the points: $f^{B_{box}} \in \mathbb{R}^{2 \times c}$,

4. $f^{key} = f^I + f^D$, $f^{key} \in \mathbb{R}^{c \times h \times w}$,
5. $f^{query} = Cat(f^{iou}, f^{mask}, f^S)$, $f^{query} \in \mathbb{R}^{(5+K) \times c}$, $f^{iou} \in \mathbb{R}^{1 \times c}$, f^{iou} is the iou token, $f^{mask} \in \mathbb{R}^{4 \times c}$, f^{mask} consists of three masks and one background token,
6. $f^{key}, f^{query} = MaskDecoder(f^{key}, f^{query}, f^{pe})$, f^{pe} is the position encoding,
7. $f^{iou} = MLP(f^{qurey}[:, 0, :])$, $f^{iou} \in \mathbb{R}^3$, $f^{mask} = MatMul(MLP(f^{query}[:, 1:4, :]))$, $f^{mask} \in \mathbb{R}^{3 \times H \times W}$.

The generation of SAM pseudolabels from a singular input does not invariably yield high-quality outcomes. Consequently, we conducted a comparative analysis of the generated pseudolabels, employing both point and box representations for distinct land cover types. In scenarios involving the segmentation of continuous land cover, such as expansive grasslands or extensive concrete surfaces, the utilization of point representations confers a notable advantage over box representations. Conversely, when tasked with segmenting

TABLE I
DETAILED INFORMATION OF THE EXPERIMENTAL DATASETS

Dataset	Spatial Resolution (Unit:Meter)	Resolution (Unit:Pixel)	Satellite	Pretreatment	Coverage Area	Class Number	Quantity of Each Category	Dataset Ratio
DLRSD[40]	2	256*256	GaoFen-1 ZiYuan-3	Reducing to 224*224	Unknown	16	100	8:2
Potsdam	0.05	6000*6000	Aircraft	Cropping to 224*224	Potsdam	5	600	8:2
Pipeline[41]	0.5	512*512	DigitalGlobe Aircraft	Cropping to 224*224	Russia	1	1200	8:2
Oil Site[42]	10	256*256	Sentinel-2	Reducing to 224*224	Russia	1	1200	8:2

numerous aircraft, vehicles, or buildings, the application of box representations consistently delivers superior results. Our experimental findings underscore the robustness and specificity of this conclusion.

Therefore, in order to harness the power of SAM for optimizing pseudolabels, we introduce a sophisticated Model Connection Module. In the second phase, we derive a probability distribution matrix for the text descriptions' corresponding categories, mirroring the dimensions of the image. Regions with higher confidence in the matrix exhibit greater alignment with the text descriptions. Exploiting this characteristic, we quantify the number of connected regions within the confidence matrix. For instances where the count falls below three, we compute the coordinates of each region's central point; otherwise, we determine the coordinates of the top-left and bottom-right corners of every connected region. This information is encoded as segmentation cues for SAM to decipher the remote sensing image, and all encodings undergo processing by the SAM lightweight masker to yield the ultimate pseudolabels.

This process can be described by (9). Within the binary matrix $M_c \in \mathbb{R}^{h\times w}$, we detect interconnected regions and encapsulate them within minimal bounding rectangles. These delineated boxes serve the purpose of masking the affinity weights matrix, A. Each pixel is then refined by associating itself with semantically analogous pixels based on the masked affinity weights. This procedural framework can be formalized as follows:

$$M_c^{\text{aff}} = B_c \odot A^t \cdot \text{vec}(M_c). \tag{9}$$

In the expression, $B_c \in \mathbb{R}^{1\times \text{hw}}$ represents the box mask obtained from the CAM of class c, $\odot$ denotes the Hadamard product, t denotes the number of refining iterations, and $\text{vec}(\cdot)$ refers to the vectorization of a matrix. Simultaneously, we performed matrix multiplication between the mask generated in the second stage and the mask generated by SAM to obtain the final segmentation mask.

IV. EXPERIMENTS

A. Datasets

In this work, we used four semantic segmentation datasets, including two public datasets, ISPRS Potsdam and DLRSD [40]. To verify the advantages and limitations of the proposed method, we also used two custom datasets: Pipeline [41] and Oil Site [42]. The detailed information of all datasets is provided in Table I. 80% of the images in each dataset are used for training, and 20% of the images are used for testing. And the validation dataset is only used for evaluating the model and does not participate in weight optimization. Moreover, to organize the dataset into image-text pairs, we found that specific descriptive details regarding the class within the prompt had minimal impact on the final results when the prompt contains the name of the target class. We established "A remote sensing image with {class}" as a universal prompt format to precisely capture the context for CAMs.

1) Custom Datasets: In this study, we have built a dataset for the over-ground petroleum and gas pipelines [41] using remote sensing images obtained from the WayBack Imagery. The Pipeline dataset is constructed using WayBack images from June 8, 2022, encompassing the Arctic region of Russia, with the total petroleum and gas pipeline length approximating 917 km. Each image within the dataset spans 400 m in length and 400 m in width, covering a total area of 1.6 km^2. We also used the Northeast Petroleum University oil well sites version 2.0 (NEPU-OWS V2.0) [42] based on the 10-meter resolution images from Sentinel-2 imagery. In our work, a well site refers to the bare ground which contains drilling equipment. The Oil Site dataset covers an area of 263 km. A geographic information processing software named ArcGIS was used in the annotation task. These two categories are rarely found in current semantic segmentation datasets. Basic experiments on these two datasets were reported in our articles [41], [42] for reference.

2) Public Datasets: We preprocessed the publicly available multiclass land cover datasets, Potsdam and DLRSD [40], by distinguishing different classes within each label image and converting them into binary classification labels. For the Potsdam dataset, both the images and labels were synchronously cropped into small patches 224 × 224 pixels, with pixels above a 0.3 proportion threshold being precisely annotated with their corresponding classes.

B. Experimental Setup

The computational resources utilized in our experiment include two NVIDIA Corporation GP102GL Tesla P40 GPUs and Intel Xeon Silver 4214 CPUs. During the fine-tuning process across all datasets, we conducted training for 1–6 epochs. Our selection of the model relied on observing the lowest contrastive loss. The fine-tuning parameter "r" was

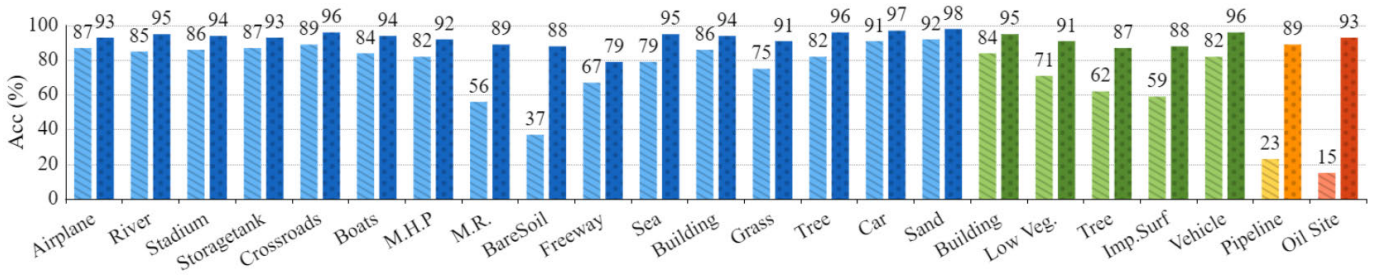


Fig. 5. Classification accuracy of CLIP for different classes before and after fine-tuning. The blue one is the DLRSD dataset, the green one is the Potsdam dataset, and the orange and red ones represent two custom datasets. The bars decorated with diagonal lines represent the accuracy before fine-tuning, and the bars decorated with dots represent the accuracy after fine-tuning.

specified as 8, "LoRA alpha" was set to 16, and a dropout rate of 0.1 was applied to the LoRA parameter layer. The target layer encompasses all aspects of the multihead attention mechanisms, two fully connected layers of the MLP, as well as the concluding patch embedding, visual projection, and text projection layers. According to our parameter formula calculation, the fine-tuning parameters constitute between 2% and 4% of the total parameters in the pretrained CLIP ViT-B16 model.

C. Evaluation Indicators

To quantitatively evaluate the footprint extraction accuracy, we select the following comprehensive metrics: Accuracy (Acc), intersection-over-unions (IoUs) score, and $F1$-Score. We also evaluated the zero-shot capability of fine-tuned CLIP using (Acc). They are defined as

$$\text{Precision} = \frac{TP}{\text{TP} + \text{FP}} \tag{10}$$

$$\text{Recall} = \frac{TP}{\text{TP} + \text{FN}} \tag{11}$$

$$F1 - \text{Score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \times 100\% \tag{12}$$

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100\% \tag{13}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \times 100\%. \tag{14}$$

In semantic segmentation, true positive (TP), true negative (TN), false positive (FP), and false negative (FN) represent the classification results of the model at the pixel level. TP means that the model correctly classifies the pixel as the target category, TN means that the model correctly classifies the pixel as the nontarget category, FP means that the model incorrectly classifies the pixel as the target category, and FN means that the model incorrectly classifies the pixel as the nontarget category. $F1$-Score is a comprehensive metric that combines precision and recall, reflecting the model's accuracy in prediction and coverage. A high accuracy rate means that the model has a high proportion of correctly classified pixels overall, and Acc is the overall proportion of correct model predictions.

D. Impact of Fine-Tuning on CLIP's Zero Shot Capability

As shown in Fig. 5, we conducted a zero-shot experimental evaluation of the classification ability of each category before and after CLIP fine-tuning. The images employed for fine-tuning have already produced promising results with our methodology. Looking ahead, as the volume of remote sensing imagery continues to expand, the prospect of retraining CLIP from scratch could be considered, but in practical terms, this approach is often prohibitively expensive. Fine-tuning emerges as a viable alternative, making it essential to explore whether this process can effectively enhance the zero-shot capabilities of existing CLIP models. In our experiments, we allocated 80% of each dataset for fine-tuning, with the remaining 20% set aside as novel data to assess the efficacy of this approach. We fine-tuned the ViT-B/16 CLIP model and evaluated the improvement in zero-shot performance by analyzing the classification accuracy across each category, both before and after fine-tuning. The quantitative results are shown in Fig. 5. We randomly selected images from the test set to obtain qualitative results, which are shown in Fig. 6.

As observed from Figs. 5 and 6, fine-tuning consistently enhances the classification accuracy of CLIP across all remote sensing categories, demonstrating a positive corrective effect on CLIP's classification outcomes. This impact is particularly pronounced in the custom datasets. The custom dataset (orange and red bars) shows significant improvements in accuracy for pipelines and oil sites after fine-tuning. Oil site shows a significant improvement from an accuracy of 15% before fine-tuning to 93% after fine-tuning, indicating that fine-tuning was particularly effective for this class. However, some categories such as rivers, stadiums, and cars have accuracies above 85% before and after fine-tuning. One possible explanation for this trend is that the categories in the custom datasets are either absent or underrepresented in CLIP's pretraining data, whereas the limited improvement seen in the public datasets may be attributed to the fact that CLIP's pretraining data already includes a substantial amount of data from these publicly available categories.

E. Quantitative Evaluation and Comparison

In order to validate the superiority of our proposed approach, we conducted comparative experiments involving cutting-edge methods [12], [13], [13], [27] leveraging CLIP or SAM, alongside traditional WSSS techniques [29], [43], [44].

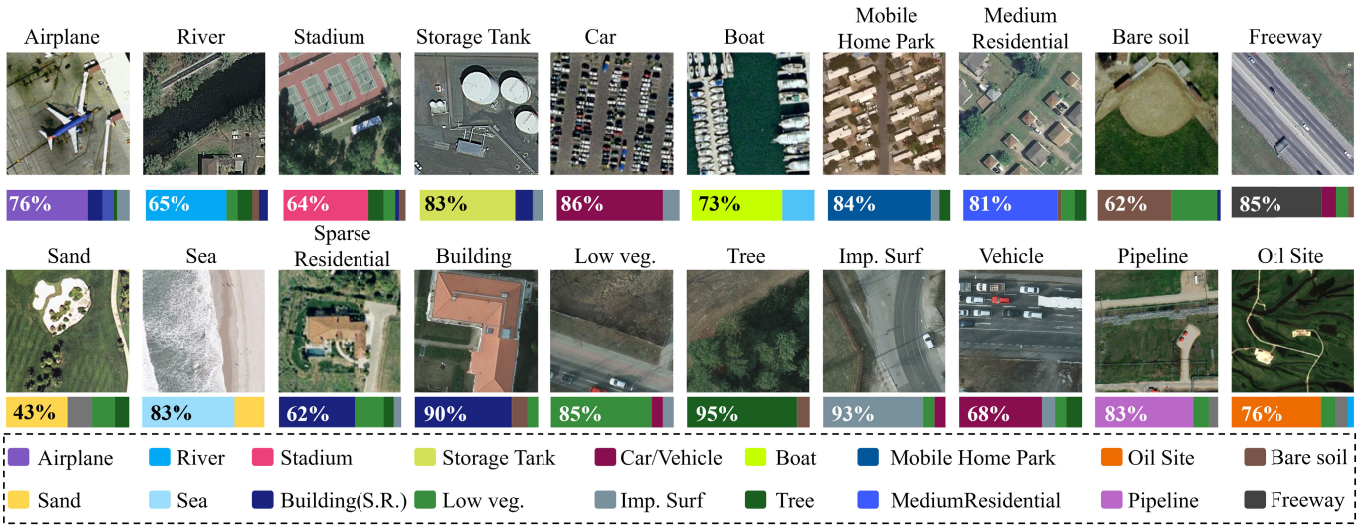


Fig. 6. Randomly select each category to test the accuracy of CLIP's zero sample classification ability.

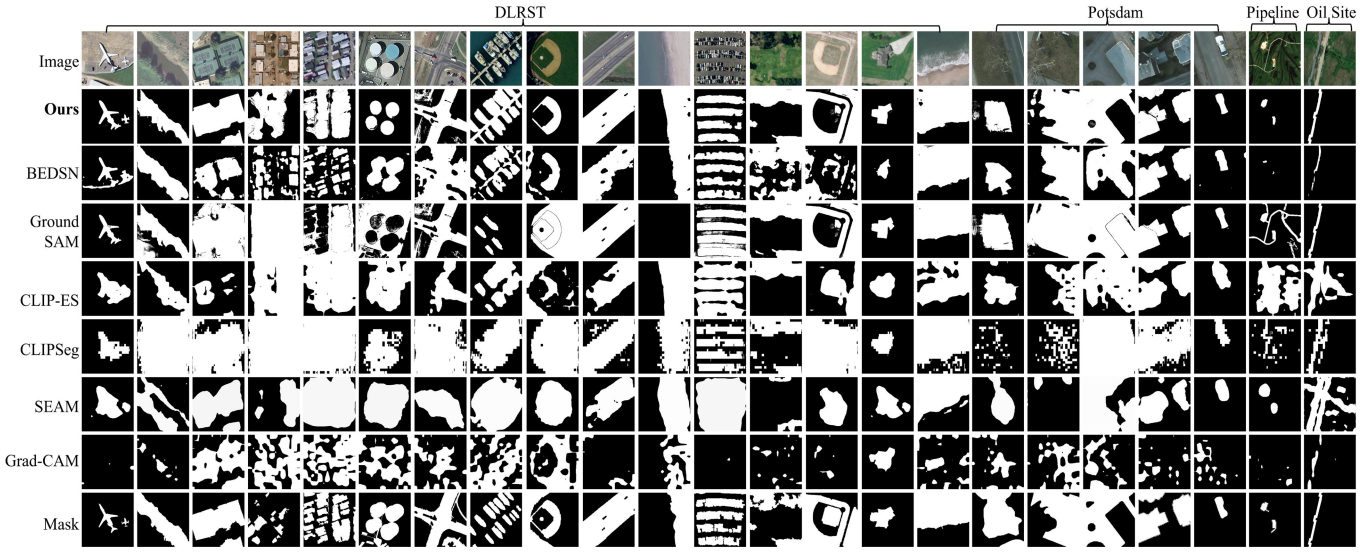


Fig. 7. Qualitative results obtained through different methods on various datasets.

We record the performance of each method (model fitting time and number of training parameters) and its average segmentation level on each dataset in Fig. 1. Compared with other methods, our method has the following advantages.

1) Reduced Background Noise: As shown in Table II and Fig. 7, since our method combines CLIP and SAM, compared with GroundSAM which only uses SAM, our advantage is that there is almost no background noise. Compared with methods such as CLIP-ES that only use the initial weights of CLIP, our method can accurately locate other categories outside the CLIP pretraining dataset after fine-tuning. Compared with other weakly supervised methods such as SEAM [43] and BEDSN [44], our advantage is that we use the powerful pretraining data foundation of CLIP to eliminate background outside the dataset category. In addition, the $F1$-Score and Acc in Table III can also directly prove our method's ability to accurately locate the foreground.

2) Sharper Segmentation Boundaries: We found that due to the particularity of remote sensing images, the edges and background of remote sensing categories are often difficult to distinguish. Using traditional methods such as Grad-CAM cannot obtain accurate segmentation ranges. As shown in Fig. 7, the GroundSAM method using SAM can always obtain clear segmentation edges, so we introduced SAM to optimize the mask. Compared with other methods, our method ensures accurate target positioning by fine-tuning CLIP, and obtains sharp segmentation edges with the help of SAM, and generally obtains satisfactory masks in the four datasets. However, our method is not as good as BEDSN in segmenting medium residential and mobile home parks. We analyzed that the reason may be that the characteristics of these two classes are multiple and scattered targets of different shapes. Obtaining masks from CLIP classifiers is not as good as directly letting CNN or Transformer structure networks train image data.

3) Enhanced Learning Efficiency: As shown in Fig. 1, we compare the performance of each method. Our method obtains higher IoU on four datasets than most weakly supervised methods, and is slightly behind the fully supervised

TABLE II
EVALUATION RESULTS (IOU/$F1$-SCORE/ACC) OF DIFFERENT METHODS ON VARIOUS DATASETS

Dataset	Class	Grad-CAM [29]	SEAM [43]	CLIPSeg [13]	CLIP-ES [12]	GroundSAM [27]	VWL [3]	BEDSN [44]	Ours
DLRSD	Airplane	14.6/20.6/27.5	25/34.1/43.7	36.0/49.6/50.9	21.1/35.3/18.4	22.2/32.5/27.6	8.00/9.40/5.50	31.7/53.9/57.2	**38.5/56.6/60.3**
	River	31.1/57.3/58.9	36.8/67.2/71.6	39.6/71.1/61.4	56.3/86.7/76.4	38.3/45.9/42.4	**72.9/85.7/90.7**	64.1/81.2/75.4	66.3/73.6/80.5
	Stadium	54.4/63.6/50.8	72.0/93.6/82.4	32.7/58.1/52.9	72.1/76.6/81.3	31.2/47.9/52.2	69.7/86.9/89.4	72.5/64.8/73.2	**82.3/94.4/87.5**
	Storagetank	20.2/44.8/20.4	38.4/52.9/37.6	36.2/62.4/71.9	40.1/61.3/73.9	24.7/44.5/33.7	51.3/61.3/56.4	51.4/69.3/51.1	**57.5/63.4/78.5**
	Crossroads	43.8/50.7/65.5	49.9/49.8/47.2	60.4/75.2/81.4	62.4/88.3/90.2	58.3/66.4/78.7	50.1/62.6/55.3	47.1/57.4/52.1	**67.2/74.3/62.3**
	Boats	13.1/21.4/10.7	45.0/61.4/66.3	**49.7/55.4/68.9**	41.2/54.8/36.9	31.0/55.2/51.4	52.1/67.5/55.5	46.9/86.4/84.3	48.1/75.8/79.3
	Baresoil	23.5/25.1/31.5	40.0/79.4/80.6	28.3/30.2/33.5	48.2/82.5/89.0	45.1/75.9/60.5	36.8/43.6/59.7	53.8/73.4/67.9	**64.7/84.5/78.2**
	Freeway	43.9/51.1/64.1	61.0/74.0/84.3	**70.4/81.5/85.4**	69.7/76.9/74.3	70.3/81.8/74.3	50.1/63.4/62.3	56.4/73.8/64.4	69.5/78.5/85.4
	Sea	39.4/42.5/44.7	43.4/56.4/61.1	77.9/85.5/73.1	68.3/70.4/83.0	68.2/64.0/77.3	58.4/63.5/59.3	72.3/86.9/84.5	**81.6/93.5/91.5**
	Sand	26.6/32.0/36.5	33.9/42.2/35.3	41.3/53.6/47.7	78.2/83.2/72.9	31.5/43.5/35.8	41.0/74.4/71.3	79.2/83.6/72.1	**81.2/94.1/84.3**
	Building	13.5/15.6/25.4	40.6/43.6/54.7	52.8/64.0/66.4	47.0/73.4/79.1	45.3/71.0/67.4	40.6/76.5/70.2	51.3/65.5/53.5	**58.7/63.7/70.8**
	Grass	37.6/43.5/38.6	53.5/63.4/63.2	64.6/73.4/62.4	74.6/77.5/84.6	74.7/84.5/79.3	50.6/68.7/59.7	49.2/70.6/75.3	**74.9/86.5/82.8**
	Tree	44.5/51.3/61.7	40.0/42.4/40.5	37.9/34.4/43.2	33.8/46.3/53.5	48.2/60.2/67.4	59.0/66.5/64.3	55.3/64.5/54.6	**80.2/92.2/87.4**
	Car	24.8/38.7/41.1	43.7/54.6/48.7	55.7/65.4/63.3	56.2/66.4/63.9	0.40/21.3/1.20	16.1/34.5/15.3	61.2/74.5/77.6	**66.4/73.9/80.6**
	M.H.P	4.60/16.6/10.7	32.8/43.5/25.1	38.7/57.8/42.2	40.8/50.4/55.7	36.6/41.2/50.6	24.6/36.5/41.2	**45.7/63.4/65.1**	31.9/56.3/52.2
	M.R.	34.7/47.7/34.3	33.8/53.5/38.9	26.3/41.8/51.9	36.4/54.6/51.4	16.6/27.1/31.8	24.6/40.1/42.7	**46.9/60.1/64.5**	39.2/53.7/55.9
Potsdam	Building	42.8/61.5/66.4	39.1/45.3/57.8	47.3/64.9/61.1	52.7/50.2/53.5	31.4/53.3/59.4	49.5/67.4/72.6	58.1/62.3/70.2	**60.5/72.3/85.4**
	Low veg.	13.6/25.7/33.1	39.4/54.3/59.2	27.6/51.7/58.2	36.3/63.6/73.8	1.70/10.6/11.4	**60.5/82.5/85.1**	42.3/70.6/81.1	46.5/78.4/86.6
	Tree	33.6/45.3/56.7	33.3/43.4/56.0	23.4/36.5/37.8	39.8/44.7/52.5	11.0/27.9/23.4	**62.3/73.5/85.4**	49.3/82.1/87.2	47.9/74.0/84.2
	Imp. surf	24.9/38.7/33.0	35.3/48.1/47.7	28.6/34.4/35.1	26.2/32.4/35.4	32.4/46.5/39.8	61.5/76.3/62.3	54.9/65.3/52.3	**63.7/73.0/83.4**
	Vehicle	23.1/32.8/37.8	36.0/48.7/51.6	28.6/30.6/35.7	21.0/27.7/31.8	41.9/64.7/55.7	37.4/51.1/40.1	**56.8/68.7/73.2**	53.4/62.7/67.3
Pipeline	Pipeline	4.60/15.4/7.60	21.7/35.3/46.1	20.4/43.2/27.8	9.9/24.3/15.4	17.5/23.4/34.6	37.5/46.5/42.2	30.7/43.7/32.1	**43.8/79.1/84.9**
Oil Site	Oil Site	2.20/13.2/3.20	21.4/43.2/37.6	10.3/13.2/23.1	10.1/21.2/34.3	23.7/31.2/45.3	24.2/34.2/44.7	39.6/54.3/43.2	**45.4/86.1/82.1**
Average		26.7/37.1/37.4	39.8/53.4/53.7	40.6/53.6/53.7	45.3/58.6/60.0	34.8/48.7/47.8	45.1/59.6/57.8	52.9/68.5/65.7	**59.5/75.6/77.8**

method. It is worth noting that our method only needs to train 2.2M parameters, which is one-quarter of the other weakly supervised methods with the least parameters. The small number of parameters in our method is the key factor for the short training time. It only takes about 10 min for us to complete fine-tuning, while other methods take about 100× longer. Although CLIP-ES [12] and GroundSAM [27] directly use CLIP and SAM without any additional training, their performance is constrained by the limitations of the CLIP pretraining dataset, which affects their evaluation results.

F. Ablation Study

1) Effectiveness of TBBS Module: We divide the TBBS process into two stages: TBBS-1 and TBBS-2. In the TBBS-1 stage, foreground and background category descriptions are provided to Grad-CAM, allowing the extraction of the foreground category mask using the CLIP classifier. In the TBBS-2 stage, the weight matrix from CLIP's MHSA is utilized to refine and complete the mask, which is then denoised and smoothed. We document the quantitative results for each dataset at each stage in Table III, while the qualitative results for each dataset at each stage are retained in Fig. 8.

We concluded that, during the TBBS-1 stage, while certain categories such as Airplane did not generate effective masks, they successfully resisted interference from background categories. This outcome substantiates the efficacy of utilizing CLIP to derive masks through text prompts for both foreground and background categories. Moreover, in the TBBS-2 stage, we observed a significant expansion in the area corresponding to various categories within the masks, notably for categories like impervious surface. This finding affirms that leveraging the MHSA weight matrix effectively enhances the accurate coverage of the foreground within the mask.

TABLE III
QUANTITATIVE EVALUATION OF EACH STAGE OBTAINED FROM ABLATION EXPERIMENTS ON FOUR DATASETS

TBBS-1	TBBS-2	Fine Tune	SAM-Box	SAM-Point	DLRSD			Potsdam			Pipeline			Oil Site		
					IoU	*F1-Score*	*Acc*	*IoU*	*F1-Score*	*Acc*	*IoU*	*F1-Score*	*Acc*	*IoU*	*F1-Score*	*Acc*
✓					30.1	41.6	38.7	21.9	31.6	44.7	2.80	10.2	14.3	4.30	10.2	5.10
✓	✓				52.2	59.3	49.1	39.4	46.2	53.3	7.80	16.6	18.3	10.5	10.4	9.40
✓	✓	✓			58.6	64.3	61.0	44.2	68.7	59.4	21.6	55.6	47.2	32.3	75.3	80.1
✓	✓	✓	✓		**63.0**	75.9	76.1	**54.4**	72.1	81.4	37.7	51.4	60.3	**45.4**	86.1	82.1
✓	✓	✓		✓	60.1	63.2	62.7	49.3	66.3	73.9	**43.8**	79.1	84.9	11.2	29.2	34.0

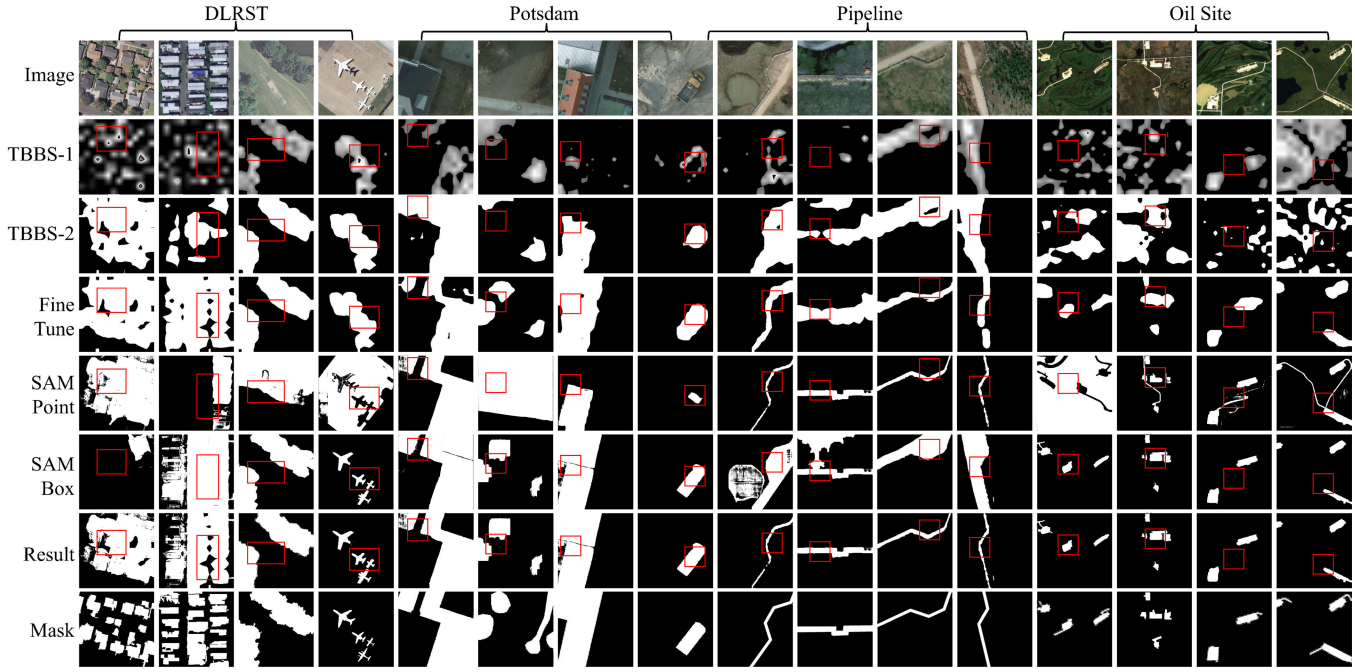


Fig. 8. Qualitative analysis of four datasets through ablation experiments. The red rectangle in the picture indicates the optimized details.

Furthermore, in the TBBS-2 stage, we observe a significant increase in the average IoU of categories, along with a significant increase in the area in the mask corresponding to these categories. This verifies that using the MHSA weight matrix can effectively increase the correct coverage area of the foreground in the mask.

Finally, we observed that the masks for certain categories, such as Pipeline and Oil Site, did not exhibit substantial improvement during the TBBS stage. These masks were either marred by significant background noise or failed to accurately correspond to the intended categories. Our analysis suggests two potential reasons for this outcome: first, CLIP may struggle to accurately classify these target categories; second, the presence of more than four categories within the image, coupled with overlapping features between background and foreground categories, poses challenges for achieving precise segmentation using weakly supervised methods.

2) Effectiveness of Fine-Tuning: We employed LoRA fine-tuning on CLIP using the training set, followed by the application of TBBS to extract masks from new data using the fine-tuned CLIP model. Both qualitative and quantitative results were obtained. Analyzing the qualitative results reveals a noteworthy improvement in IoU for the custom datasets Oil Site and Pipeline, attributable to fine-tuning CLIP. Furthermore, the qualitative results demonstrate that the masks for the Oil Site and Pipeline categories, extracted from the fine-tuned CLIP model using TBBS, more accurately localize the targets while significantly reducing background noise compared to the unfine-tuned CLIP. Considering the experimental conclusions regarding the impact of fine-tuning on CLIP's zero-shot capabilities, we can infer that fine-tuning enhances CLIP's classification performance for unfamiliar categories. Consequently, integrating the TBBS module with the fine-tuned CLIP can effectively improve the accuracy of masks for categories previously unfamiliar to CLIP.

3) Effectiveness of ML Module: Upon completing the TBBS process and fine-tuning, we observed that although target categories such as building and airplane were correctly identified and covered in the masks, the edges of these targets were not accurately delineated. To address this issue, we propose introducing SAM. SAM requires segmentation prompts such as points, boxes, or scribbles. After generating the mask, we calculate the connected regions, determine the coordinates of the upper-left and lower-right corners of these regions,

and provide this data as box prompts to SAM. Additionally, we calculate the center point of the box and use this coordinate data as a point prompt for SAM.

We applied these two types of prompts across all classes. As shown in Fig. 8 and Table III, different prompts are better suited to remote sensing categories with distinct characteristics. For classes with a single connected area, such as impervious surface, pipeline, and grass, using points as SAM prompts yielded high-quality masks. Conversely, for classes with discrete connected areas, such as oil site, airplane, and vehicle, box prompts were more effective in producing masks with reduced noise.

Finally, we multiply the mask generated by SAM with the masks obtained by the first two methods to obtain the result shown in Fig. 8. This approach had a dual benefit: SAM enhanced the mask's edge accuracy, making the segmentation of target categories neater, and the multiplication process helped mitigate SAM's tendency to generate noise in the masks, preserving mask quality. Experimental results confirmed that SAM improved the IoU across nearly all classes, validating its effectiveness in this method.

V. Conclusion

In our research, we have devised an RS-TextWS-Seg framework and substantiated through experimentation that its most significant advantage lies in the rapid generation of high-fidelity pseudolabels aimed at providing sharp and immaculate boundaries. This framework has shown promising potential in enhancing CLIP's discerning and discriminative abilities for classifying unfamiliar objects from remote sensing images. In the proposed RS-TextWS-Seg architecture, the TBBS module within RS-TextWS-Seg effectively suppresses background noise by categorically decluttering foreground and background elements, while leveraging the weight matrix of MHSA to expand the coverage area of pseudolabels. Furthermore, the ML module in RS-TextWS-Seg is geared toward refining the edges of pseudolabels, with empirical evidence demonstrating the advantages of employing SAM for distinct categories of remote sensing entities. Nevertheless, it is imperative to acknowledge the inherent limitations of our proposed methodology. Specifically, RS-TextWS-Seg's segmentation capabilities on densely distributed and scattered objects do not surpass those of supervised or weakly supervised methods trained from scratch. Looking ahead, our future endeavors will focus on exploring a weakly supervised framework capable of swiftly generating pseudolabels for densely distributed objects from remote sensing images.

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Zhibao Wang (Member, IEEE) received the Ph.D. degree in science from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2013.

He is currently an Associate Professor with Northeast Petroleum University, Qinhuangdao, China. His research interests include remote sensing image analysis, energy big data, and oil and gas data governance.



Huan Chang received the B.S. degree from the School of Computer Science and Technology, Chongqing University of Posts and Telecommunications, Chongqing, China, in 2022. She is currently pursuing the M.S. degree with the Computer Technology, School of Computer and Information Technology, Northeast Petroleum University, Daqing, China.

Her major research interests include remote sensing image analysis and remote sensing big data.



Lu Bai (Member, IEEE) received the Ph.D. degree in electronic engineering from the University of Kent, Canterbury, U.K., in 2014.

She is currently a Senior Lecturer with the School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast, Belfast, U.K. Her research has been supported by Innovate U.K., Intertrade Ireland and Industry. She has authored or co-authored more than 50 peer-reviewed publications in reputable journals and conference proceedings and a book chapter. Her recent research interests lie in the application of remote sensing technologies and geospatial analysis for environmental monitoring including the land use changes, and climate change impacts.



Liangfu Chen received the Ph.D. degree in geography from the Institute of Remote Sensing and Geographical Information System, Peking University, Beijing, China, in 1999.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include satellite retrieval of atmospheric aerosol, greenhouse, and Earth big data.



Xiuli Bi received the B.Sc. and M.Sc. degrees from Shanxi Normal University, Xi'an, China, in 2004 and 2007, respectively, and the Ph.D. degree in computer science from the University of Macau, Macau, China, in 2017.

She is currently a Professor with the College of Computer Science and Technology, Chongqing University of Posts and Telecommunications, Chongqing, China. Her research interests include image processing, multimedia security, and forensic.